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1. Title:

“How Much Synchronization Do We Need? Characterizing the Timing-Offset and Bandwidth Robustness of Domain-Invariant in the Wi-Fi CSI Indicators”

Category: Research in Wifi Sensing

Problem / System Being Addressed:

We will be intending to study CSI-based Wi-Fi sensing systems, specifically domain-invariant motion indicators such as Body-Velocity Profile (BVP). The problem we address is that the robustness of these sensing methods under synchronization errors and changes in available bandwidth is not well characterized. All the existing methods have primarily been developed and evaluated using relatively narrowband CSI under synchronized conditions but as wireless sensing systems move toward wider bandwidths and emerging sensing standards, timing/synchronization errors and bandwidth variations may alter the CSI and consequently affect sensing and cross-device generalization performance, which is what we are trying to attempt to head into in the upcoming milestones.

2. Project Summary:

Almost all the Wi-Fi sensing uses Channel State Information (CSI) to infer human activities and motion from changes in wireless signals. Although existing CSI-based sensing methods can achieve strong performance under controlled conditions, we still have limited understanding of how well they generalize under different wireless conditions. In particular, many existing methods have been developed and evaluated using narrowband CSI and relatively well-synchronized commodity Wi-Fi devices but as wireless sensing moves toward wider bandwidths and emerging standards such as IEEE 802.11bf, we expect synchronization errors and changes in available bandwidth to become important factors affecting signal fidelity and sensing performance.

In this project, we as a group will try to investigate the robustness of domain-invariant CSI motion indicators, focusing primarily on Body-Velocity-Profile (BVP)-style processing, using a public CSI dataset such as Widar 3.0, or other SOTA driven source and we will first establish a baseline sensing performance and then introduce controlled synchronization/timing offsets and bandwidth reductions to the CSI data. We will evaluate how sensing and cross-device generalization performance changes as these perturbations increase in environment or in other terms.

Our primary contribution to attend will be an empirical sensitivity analysis and characterizing the relationship between synchronization error, bandwidth and sensing accuracy with the aim of identifying a potential failure threshold. If we observe a significant degradation, we will further investigate a lightweight signal-correction technique and evaluate its ability to recover the lost performance. Through this work, we aim to connect ML based Wi-Fi sensing with fundamental wireless-communication concepts such as synchronization, bandwidth, channel behavior and signal fidelity.

3. Key Objectives

- Study the effect of timing errors on BVP. Add different levels of timing/synchronization error to Widar 3.0 CSI data and measure how BVP classification accuracy changes.
- Study the effect of bandwidth reduction on BVP. Reduce the available CSI bandwidth/subcarrier count and measure the resulting change in BVP accuracy.
- Identify the failure point. Find the timing-error and bandwidth levels at which BVP performance starts to decrease significantly. This point will be treated as the knee/failure point.
- Explore a lightweight correction method. If significant degradation is observed, develop and test a simple phase/timing correction method to check whether some of the lost accuracy can be recovered.

4. Expected Outcomes

- A timing-error vs. BVP accuracy sensitivity curve.
- A bandwidth vs. BVP accuracy sensitivity curve.
- Identification of the knee/failure point for timing error and bandwidth reduction.
- A simple correction method, if feasible, with a comparison of uncorrected vs. corrected BVP performance.
- A reproducible experimental setup using public Widar 3.0 CSI data, without requiring new hardware.
- A final conclusion on whether timing/synchronization errors and bandwidth reduction significantly affect the generalization performance of BVP, and under what conditions BVP starts to fail.

5. Data sets:

For this project, we will be using the WiFi-based Activity Recognition Dataset (Widar 3.0) [9] as our primary dataset. Widar 3.0 is a Wi-Fi sensing dataset containing Channel State Information (CSI) measurements and it is collected using Intel 5300 wireless Network Interface Cards (NICs). The dataset includes measurements across different environments, users, postures, orientations and activity/gesture classes, making it suitable for evaluating the robustness and generalization of CSI-based Wi-Fi sensing methods.

The raw CSI data will serve as the basis for our experiments and we will establish a baseline using the original CSI data and then create controlled perturbations by introducing different levels of synchronisation/timing error and reducing the available bandwidth or number of CSI subcarriers. For each experimental condition, we will process the resulting CSI using the selected BVP-based sensing pipeline and measure the corresponding classification and cross-device generalisation performance.

By comparing the performance across increasing levels of synchronisation error and bandwidth reduction, we aim to characterize the degradation and identify a potential knee or failure threshold.

The original Widar 3.0 dataset will be obtained through IEEE DataPort [9]. The experimentally generated perturbed datasets, preprocessing procedures, and data-processing scripts will be documented and preserved in the project's GitHub repository to support reproducibility.

Dataset Source: IEEE DataPort — *WiFi-based Activity Recognition Dataset* [9].
DOI: 10.21227/7ZNF-QP86.

6. GitHub Repository

GitHub Repo:
g4-research-wifi-sensing

Github Clone Link
https://github.com/dhruv-codes-pixel/g4-research-wifi-sensing

7. References

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